Haoxuan You

Sparks, NV

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Manufacturing and process engineer for lithium cell and module production, spanning pilot-line development through 24/7 mass production. Lead new line commissioning, qualification, and ramp with domestic and international OEM suppliers — process flow mapping, fixture definition, PFMEA, control plans, and validation through buy-off — and feed process-capability data back into design decisions. Scaled lithium-metal, Li-S, LFP, and solid-state cell processes from R&D to high volume. Fluent in English and Mandarin Chinese.

Education

M.S., Mechanical Engineering, Jul 2021. *Purdue University*

B.S., Mechanical Engineering, May 2019. *University of Iowa*

Professional Experience

Tesla – Sparks, NV

Process Engineer, LFP Cell Manufacturing (Dec 2025 – Present)

- Own LFP cell and module stack assembly processes for 24/7 high-volume production, holding process capability to Cpk >1.67 and reducing cycle time by 15%.
- Ramped cell assembly throughput from hundreds to tens of thousands of cells per day and brought the module stack line from 20% to 80% of design capacity at stable output.
- Led commissioning, qualification, and ramp of 7 new cell assembly stations with domestic and international OEM suppliers — process flow mapping, equipment function definition, fixture design improvement and validation, mechanical checks, dry and wet runs, PFMEA, GR&R, error proofing, MES configuration, control plan and SOP development, and process validation through OEE demonstration, buy-off, and integration into the mass-production line.
- Reduced busbar laser-weld scrap by 80% through DOE and laboratory characterization — deployed in-line vision inspection for busbar position accuracy and replaced manual cleaning with an automated process, improving weld quality while cutting cycle time and operator labor.
- Redesigned the top-cap laser welding fixture, replacing consumable PEEK tape with a screw-fastened PEEK block — eliminated tape placement variation driving weld quality variation and yield loss, and extended replacement interval from every 3 days to every 2 weeks, cutting changeover time and unplanned downtime.
- Redesigned first-article inspection sampling and metallographic sample preparation for laser welds, eliminating non-value-added steps with process-capability validation — reduced FAI time by 50% and shift-change downtime by 40% while improving throughput and line ergonomics.
- Drove yield improvement across laser welding, ultrasonic welding, hot-melt welding, and leak detection stations through structured root-cause analysis (5-Why, Ishikawa, 8D), improving overall line rolled yield by 10%.
- Configured MES data collection and in-line SPC across critical stations and deployed production dashboards tracking failures by fixture, track, batch, and occurrence pattern, plus OEE, equipment-alarm analytics, and automated Cpk trend reporting — alerting process and maintenance teams to drift early enough to correct before NG product was produced, avoiding scrap and improving yield by an average of 2% per station.

Lyten – San Jose, CA

Senior Battery Engineer (Mar 2025 – Dec 2025)

- Led cell-level mechanical and structural design for polymer-coated lithium-metal anodes, defining stack-ups, tolerances, and reliability targets for pouch-cell integration.
- Led scale-up and ramp of lithium-metal foil production (100× throughput increase) by developing extrusion, lamination, and precision rolling processes to meet tight dimensional tolerance and cosmetic quality requirements while sustaining production yield.
- Led new equipment design with vendor teams, driving manufacturability, quality, and scalability requirements into equipment specifications and applying DFM and tolerance analysis to secure yield stability at pilot scale.
- Designed and executed DOE-based mechanical and thermal validation (swelling, peel, compression, lamination), identifying key failure modes and driving corrective actions that reduced defect rates by >40%.
- Developed new manufacturing processes and optimized critical equipment and process modules for polymer-coated lithium-metal and lithium-alloy anodes, translating new component chemistries into manufacturable, scalable processes for Li-S cell platforms.
- Presented validation results and DFM recommendations to hardware leadership and executive stakeholders, informing design decisions and scale-up strategy.

Battery Engineer (Apr 2022 – Mar 2025)

- Developed and refined polymer-coating and lamination processes for lithium-metal and lithium-alloy anodes — from materials formulation and coating trials through process optimization and integration into high-volume pilot production — contributing to performance gains in Li-S cell platforms.
- Applied DFM with vendors to streamline fabrication of lithium-metal electrode processing equipment, improving throughput and first-pass yield.
- Designed and validated custom fixtures, jigs, and thermal control modules in SolidWorks, enabling precision assembly and accelerating failure-analysis cycles.
- Authored SOPs, validation protocols, and process documentation, training technicians to ensure repeatability and alignment between R&D and production.
- Automated data collection, visualization, and database management with Python and JMP, supporting DOE studies, reliability testing, and pilot-line scale-up.

Sakuu Corp. – San Jose, CA

Battery Associate Engineer (Nov 2021 – Apr 2022)

- Designed and improved cell assembly and test fixtures for solid-state and lithium-metal prototypes, ensuring precise compression and dimensional consistency.
- Conducted mechanical failure analysis on early-stage cells and implemented assembly process improvements, achieving uniform pressure distribution and improved reproducibility.

Purdue University – West Lafayette, IN

Graduate Research Assistant (Aug 2019 – Jul 2021)

- Designed and integrated a laser-based additive manufacturing (SLM) system with CNC machining for advanced metal and ceramic component fabrication.
- Conducted mechanical and microstructural characterization (hardness, fracture toughness, SEM, optical microscopy) and applied DOE to establish process repeatability.

Technical Skills & Expertise

- **Process & Manufacturing Engineering:** New line commissioning, qualification, and ramp-up; process flow mapping; PFMEA; control plans; GR&R / MSA; SPC and process capability (Cpk); OEE; DOE; 8D, 5-Why, Ishikawa; SOP authoring; error proofing; DFM, design reviews, DVP&R
- **Processes & Assembly:** Laser, ultrasonic, and hot-melt welding; leak detection; electrode coating, calendering, lamination, extrusion and precision rolling; coin, pouch, and cylindrical cell builds; module stack assembly; pilot-to-production transfer
- **Automation & Data:** MES configuration and data collection; in-line vision inspection; automated cleaning; SPC and OEE dashboards; Python, SQL, MATLAB, LabVIEW, C/C++; JMP, Tableau, ImageJ
- **Design & Analysis:** GD&T, tolerance stack-up, fixture, jig, and module design; SolidWorks (CSWP), Creo; FEA (ANSYS, Abaqus); 3D printing (SLS/SLM), machining, laser processing, thin-film deposition
- **Test & Characterization:** Metallography and weld cross-sectioning; mechanical/electrochemical failure modes; EIS, CV, GITT, PITT on Arbin, Maccor, Neware, and BioLogic cyclers; mechanical load testing, dimensional and pressure mapping, profilometry, SEM/EBSD, XRD
- **Languages:** English and Mandarin Chinese (fluent) — used daily in coordination with international suppliers

Publications

- Zhu, Y., **You, H.**, et al. (2025). 3D Composite Anode Enables the Long Cycle Life of Li-S Batteries. Accepted for presentation at the 247th ECS Meeting, Montréal, Canada.
- Meng, Y., **You, H.**, et al. (2025). Anode Design Strategies for High Energy Li-S Batteries with Prolonged Cycle Life. Accepted for presentation at the 247th ECS Meeting, Montréal, Canada.
- Zhu, Y., **You, H.**, et al. (2024). The Strategy to Tune Li Stripping/Plating Behavior in Li Alloy Anodes. Presented at the 2024 Electrochemical Society (ECS) Meeting, San Francisco, CA.
- Liu, W., **You, H.**, et al. (2021). Double-pulse laser micro sintering: experimental study and mechanism analysis aided by in-situ temperature measurements. Journal of Manufacturing Processes.
- Wang, Q., **You, H.**, et al. (2020). An innovative laser metasurface fabrication technique for flexible optoelectronic devices. Journal of Micro and Nano-Manufacturing.

Patents

- Shan, J., **You, H.**, et al. (2024). Hybrid Gel Polymer Electrolytes and Catholytes for Lithium-Sulfur Batteries. Patent WO2024118412A1. Assigned to Lyten, Inc.
- Meng, Y., **You, H.**, et al. (2025). Lithium-Sulfur Batteries. Patent WO2025085743A1. Assigned to Lyten, Inc.
- Bugga, R., **You, H.**, et al. (2024). Lithium-Alloy Anodes for Lithium-Sulfur Batteries. Patent WO2024259126A3. Assigned to Lyten, Inc.